Generalization and Adaptation of MLP for News Classification

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1. Introduction

In this project, we wanted to classify news articles from multiple sources using multiple datasets into one of five topics: Politics, Sports, Business, Technology, or Entertainment. We trained models using Data A and evaluated whether models trained on one news source could generalize to both unseen Data A articles and Data B articles. We also studied how regularization affected performance and later examined what happens when the classes change in Data C. Since machine-learning classifiers require numerical input rather than raw text, we represented each news article using term frequency inverse document frequency (TF-IDF) features. For each example we concatenated the article title and description the converted the text into a unigram TF-IDF vector. TF-IDF gives greater weight to words that are important in a specific article but less common across all articles. This allows the models to use words that are related to the article specifically while reducing the impact of common words across multiple articles. We compared two models which were logistic regression (LR) and multi-layer perceptron (MLP. LR provides a linear baseline that predicts a topic from a weighted combination of TF-IDF features. The MLP uses two hidden layers with ReLU activation functions which allow it to learn nonlinear relationships. We selected regularization settings using a validation split from Data A only then refit the selected models on all available Data A training data. Then Data A and B were used for model selection and training.

2. Part 1: Exploratory Data Analysis

For this part, we wanted to make sure to clean the data. Cleaning the data helps to make sure that the data is accurate for the model to learn and it keeps the model from encountering any errors that may prevent it from correctly learning the model. For this part we only cleaned dataset_a_train. The total number of examples before data cleaning was about 7500. After dropping duplicate examples and examples with null values, the data decreased to about 7,453 examples. The 47 examples that were removed all belonged to the class "Technology." Therefore, there were 1500 examples for each class and Technology had about 1,453. Duplicate examples could give more weight to certain examples on accident while null examples can confuse the model by taking in a Title or Description instead of both. For this project, each token consists of exactly one word. The description and title are split according to spaces to account for this. We calculated the distributions of both the Title and Description using Matplotlib's boxplot functions.

With some observation, the median token length for a Title is about 10 and the median token length for a Description is around 20. It's logical for a Title to have less words than a Description. As for the range of values, the Title could range from 1 to 25 while a Description could range from 1 to around 160. Clearly the Description takes on 5 times the values as the Title which could make the model more favorable towards Descriptions since there's also 5 times the data for the model to work with. The Title length distribution is roughly symmetrical while the Description distribution is mainly left skewed.

Below are some representative examples from each class:

1) **Class 1**: Politics
   Title: Movement Against Juvenile Court Fees Runs Into Resistance
   Description:
   In almost every state and the District of Columbia, the minors who appear in the roughly one million juvenile court cases
   Analysis: Mentions the words court and cases. Although cases can be attributed to different meanings.
2) **Class 2**: Sports
   Title: Lindsey Vonn Wins Bronze In What's Likely Her Last Olympic Downhill Race
   Description:
   "I have no regrets; I'm really happy," she said.
   Analysis: Mentions phrases such as the Olympics and bronze.
3) **Class 3**: Business
   Title: 10 Habits of Ultra-Likeable Leaders
   Description:

*. The authors are listed alphabetically.

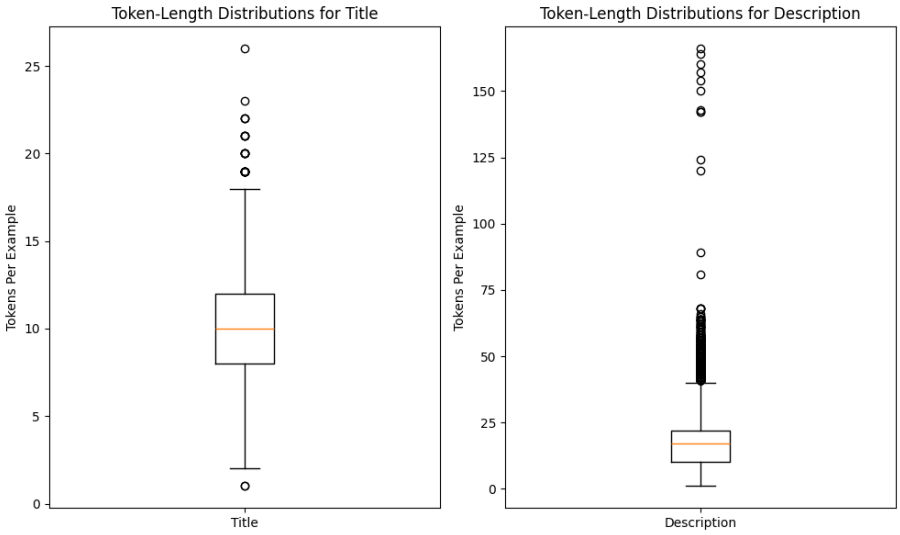


Figure 1. Token Length Distribution for Title and Description fields

Becoming a more likable leader is completely under your control, and it's a matter of emotional intelligence (EQ). Unlike innate, fixed characteristics, such as your intelligence (IQ), EQ is a flexible skill that you can improve with effort.

Analysis: Mentions phrases such as leader, intelligence, skill, and other business-like phrases.

4) **Class 4**: Technology
Title: Apple Accelerates Development Of Its Electric Car
Description:

Apple Inc. is accelerating efforts to build an electric car, designating it internally as a "committed project" and setting

Analysis: Mentions phrases such as Apple, an electric car, and "committed project" which are all tech related words.

5) **Class 5**: Entertainment
Title: 5 Netflix Things To Watch If You Can't Wait For The Royal Wedding
Description:

If the upcoming royal wedding is very much your thing, this is for you.

Analysis: Mentions the phrases "Netflix" and "Watch"

Then we also wanted to check to see if there was any vocabulary in the data that could made the model predict the classes incorrectly. We noticed that there was a Star Wars example classified as Technology when there were phrases mentioned like "BB-8" and droid. The Star Wars example should have been classified as Entertainment and the model may use the incorrect data to classify most Star Wars-related examples as Entertainment. There are also some words that have multiple meanings. For example, a bank could refer to the area near a river, or it could mean a place where people deposit money into. The model may get confused with certain important words and classify the examples incorrectly.

3. Part 2: In-Domain Performance on Dataset A

We start by comparing the performance of the multilayer perceptron (MLP) with logistic regression (LR) on Dataset A. The training dataset was partition using stratified sampling into $6,000$ training and $1,500$ validation examples, preserving the class distribution in both sets. A raw unigram TF-IDF vectorizer was fit exclusively on the training partition. The Title and Description fields were concatenated for each article so that both headline and contextual information were available for the models. Since LR serves as a linear baseline, whereas the MLP introduces nonlinear transformations to the TF-IDF representation, comparing the two models therefore allows us to examine whether the extra nonlinear capacity of the MLP improves in-domain classification.

For the MLP, we focus on designing a stable model with sufficient capacity to capture nonlinear feature while controlling overfitting. As shown in Table 1, the MLP uses two hidden layers with 128 and 64 units, respectively. The first hidden layer transforms the high-dimensional TF-IDF input into 128 latent features, which are further reduced to 64 featues before the final output layer. ReLU activations introduce nonlinearity, while dropout and L2 regularization are used to reduce overfitting. Both models were trained using SGD with a batch size of 32 and cross-entropy loss as an optimizing. LR used

a learning rate of 0.01 while the MLP used a learning rate of 0.024, momentum of 0.9, dropout rates of $0.2 \rightarrow 0.3$, and $\mathcal{L}2$ regularization (weight decay) of 10^{-3}. Training duration was determined based on the model's validation behavior with an early-stopping patience of five epochs. After these decisions were fixed, the vectorizer was refit on all $7,500$ examples in the Dataset A training set, and both models were re-trained using their selected training durations before a single evaluation on the 2,500-example Dataset A test set.

TABLE 1. INITIAL CONFIGURATIONS FOR LR AND MLP

Model	Architecture	Optimizer	Regularization	Epochs	Vocab. Size
LR	Single Linear	SGD	None	50	19,834
MLP	$128 \rightarrow 64$ with ReLU	SGD	Dropout 0.2/0.3; $\mathcal{L}2 = 10^{-3}$	11	19,834

TABLE 2. AVERAGE IN-DOMAIN PERFORMANCES OF LR AND MLP ON TEST DATASET A

Model	Accuracy	Macro-F1
LR	0.5156	0.5171
MLP	0.5464	0.5375

Table 2 shows that the MLP achieved an accuracy of 0.5464 and a macro-F1 score of 0.5375, compared with 0.5156 accuracy and 0.5171 macro-F1 for LR. The raw improvements are 0.0308 in accuracy and 0.0204 in macro-F1. The increase in both metrics suggests that the MLP achieved better overall predictive performance as well as improved average class-wise performance. Because macro-F1 accounts for equal importance to each class, its improvement cannot be attributed simply to better performance on a more frequent class. Furthermore, each class contains exactly 500 examples in the Dataset A test set, consistent with the balanced class distribution identified during the analysis in Section 2.

TABLE 3. PER-CLASS PERFORMANCES OF LR AND MLP ON TEST DATASET A

Model	Class	Precision	Recall	F1
LR	Politics(1)	0.3952	0.4260	0.4100
LR	Sports(2)	0.5596	0.6200	0.5882
LR	Business(3)	0.5230	0.5000	0.5112
LR	Technology(4)	0.6690	0.5700	0.6156
LR	Entertainment(5)	0.4592	0.4620	0.4606
MLP	Politics(1)	0.4467	0.3100	0.3660
MLP	Sports(2)	0.5838	0.7800	0.6678
MLP	Business(3)	0.6193	0.5140	0.5617
MLP	Technology(4)	0.6603	0.6960	0.6777
MLP	Entertainment(5)	0.3978	0.4320	0.4142

Nonetheless, as shown in Table 3, the improvement is not uniform across all classes. The MLP increases the F1 score for Sports from 0.5882 to 0.6678, Business from 0.5112 to 0.5617 and Technology from 0.6156 to 0.6777. By contrast, Politics decreases form 0.4100 to 0.3660 and Entertainment from 0.4606 to 0.4142. The largest improvement occurs for Sports, where recall increases from 0.5700 to 0.6960. In contrast, the decrease in Politics F1 is largely associated with its lower recall of 0.3100 under the MLP. These results indicate that the additional nonlinear capacity of the MLP benefits some categories but fails to identify the lexical ambiguity present across all classes.

From the error analysis, we observe that LR misclassified $1,211$ of $2,500$ test articles, whereas the MLP misclassified $1,134$. A follow-up qualitative inspection of ten representative errors reveals four distinct patterns: Politics–Entertainment overlap, Business–Technology overlap, Sports–Entertainment overlap, and cases in which the description is usually short or poorly correlated to its corresponding category.

The class-wise results in Figure 2 further show that the MLP performs most effectively on Sports and Technology, with F1 scores of 0.6678 and 0.6777, respectively, whereas Politics and Entertainment remain the most difficult categories. This pattern may be explained by overlapping vocabulary among different classes and by articles whose descriptions provide limited additional information about the article's topic. For example, company-related articles can contain vocabulary associated with both Business and Technology, while stories involving public figures may contain cues associated with both Politics and Entertainment. These overlaps make the class boundaries difficult to determine using lexical TF-IDF features alone.

Summary. The MLP provides an improvement in in-domain performance over the linear baseline, with higher overall accuracy and macro-F1. However, the relatively modest increase in macro-F1 indicates that nonlinear modelling does not necessarily eliminate ambiguity between semantically overlapping classes. The class-wise and error-analysis results suggest that the remaining errors are associated with lexical overlap, weakly informative descriptions, and source-specific patterns identified during EDA.

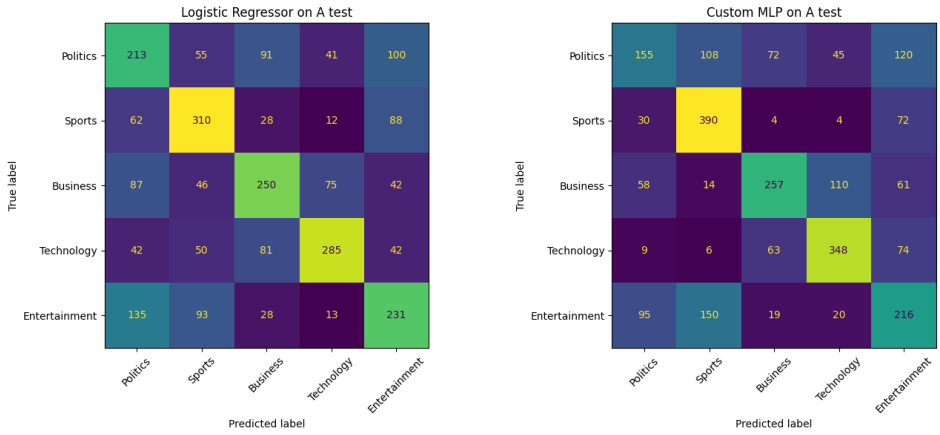


Figure 2. Confusion matrices of LR and MLP on Dataset A test set

4. Part 3: Regularization

To implement a fully functional logistic regression model, we had to test which regularization settings and which MLP settings would be most sufficient. For the MLP and LR models there were regularization settings we wanted to experiment with such as the weight decay and dropout layers. With the settings in mind, we compared the F1 scores of the parameter adjustments we made for both LR and MLP models on validation data A.

TABLE 4. REGULARIZATION CONFIGURATIONS FOR LR AND MLP

Model Name	Learning rate	$\mathcal{L}2$	Dropout
`lr-wd0`	10^{-2}	0	None
`lr-wdle-5`	10^{-2}	10^{-5}	None
`lr-wdle-4`	10^{-2}	10^{-4}	None
`lr-wdle-3`	10^{-2}	10^{-3}	None
`mlp_none`	10^{-3}	0	None
`mlp_dropout`	10^{-3}	0	None
`mlp_dropout_wd`	10^{-3}	10^{-4}	None
`mlp_strong`	10^{-3}	10^{-3}	None

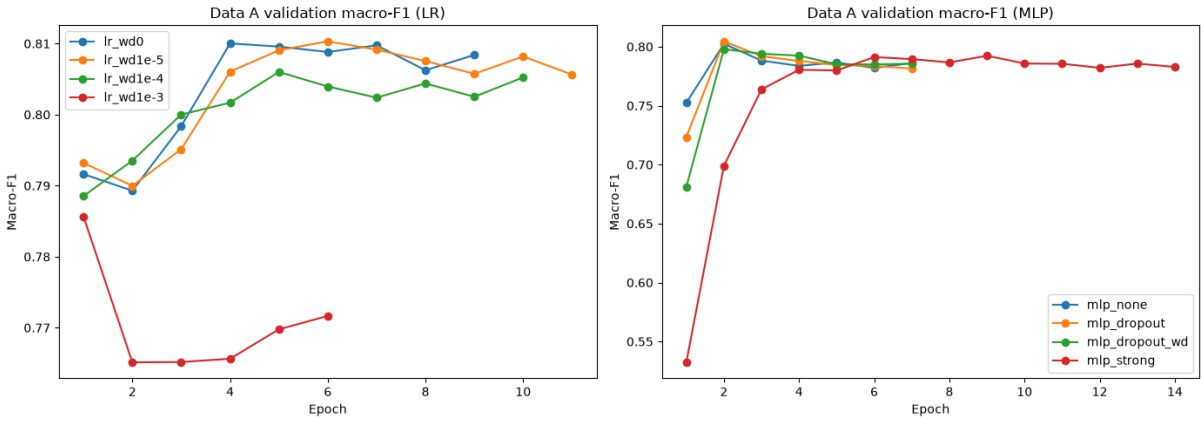


Figure 3. Macro-F1 scores of LR and MLP on Dataset A validation set

Note that the patience parameter makes it, so a few graphs don't show the macro-F1 at certain epochs if there isn't a big enough change in validation macro-F1 score. For the MLP, the `mlp_dropout` reached the highest macro-F1 score of 0.804 while for LR the `lr_wdle-5` model reached the highest macro-F1 score of 0.81. With these conclusions, we decided to select the `mlp_dropout` and `lr_wdle-5` models for the rest of the experiment. After validating which parameters worked best, we then refitted `mlp_dropout` and `lr_wdle5` to learn from training data A. There was a significant

decrease in accuracy and macro-F1 score for Dataset A train set where the LR model had 0.5648 and 0.5647 respectively and the MLP model had 0.5784 and 0.5748. As for the Dataset B test set, the LR model achieved a 0.78 accuracy and 0.78 macro-F1, while the MLP model achieved a 0.79 for accuracy and a 0.79 for macro-F1. The model performed much better on Dataset B compared to Dataset A.

TABLE 5. MACRO-F1 SCORES PER MODEL

Model Name	Macro-F1 (A)	Macro-F1 (B)
`lr-wd0`	0.5599	0.7864
`lr-wdle-5`	0.5647	0.7888
`lr-wdle-4`	0.5626	0.7597
`lr-wdle-3`	0.5472	0.5494
`mlp_none`	0.5749	0.8230
`mlp_dropout`	0.5749	0.7941
`mlp_dropout_wd`	0.5723	0.7797
`mlp_strong`	0.5521	0.7349

We also checked to see if the other model settings could produce a better macro-F1 score It turned out that the MLP settings we chose had a 10% worse macro-F1 score than the `mlp_none` settings on dataset B. It's possible that the settings we chose work best with certain types of data. For example, the best LR model for B according to micro-F1 happened to not be the best LR model for A.

TABLE 6. ACCURACIES AND MACRO-F1 SCORES OF THE BEST MODELS

Model Name	Accuracy (A)	Macro-F1 (A)	Accuracy (B)	Macro-F1 (B)
`lr-wd0`	0.5648	0.5647	0.7800	0.7889
`mlp_none`	0.5784	0.5749	0.7900	0.7941

After selecting settings using only Dataset A's validation macro-F1, we refitted the selected LR and MLP configurations on Dataset A. We used a unigram TF-IDF vectorizer on both the title and description test from Dataset A training set only. The vectorizer was then frozen and used to experiment with Dataset A test and Dataset B test. Both classifiers performed substantially better on Dataset B than on Dataset A test. Logistic regression increased from 0.5647 macro-F1 on Dataset A to 0.7888 on Dataset B. The MLP increased from 0.5749 to 0.7941 macro-F1. The MLP performed slightly better than logistic regression on both Dataset A and Dataset B, although the difference was small. We then tried to compare performance according to the classes that were determined. Table 7 were the results.

TABLE 7. PER-CLASS F1, PRECISION, AND RECALL METRICS ON DATASET A AND B

Model	Class	F1 (A)	Precision (B)	Recall (B)	F1 (B)
LR	Politics	0.379	0.938	0.542	0.687
LR	Sports	0.658	0.982	0.871	0.923
LR	Business	0.587	0.527	0.990	0.688
LR	Technology	0.683	0.981	0.643	0.777
LR	Entertainment	0.517	0.977	0.782	0.869
MLP	Politics	0.388	0.874	0.468	0.609
MLP	Sports	0.681	0.983	0.896	0.938
MLP	Business	0.580	0.556	0.986	0.711
MLP	Technology	0.702	0.931	0.746	0.828
MLP	Entertainment	0.523	0.990	0.798	0.884

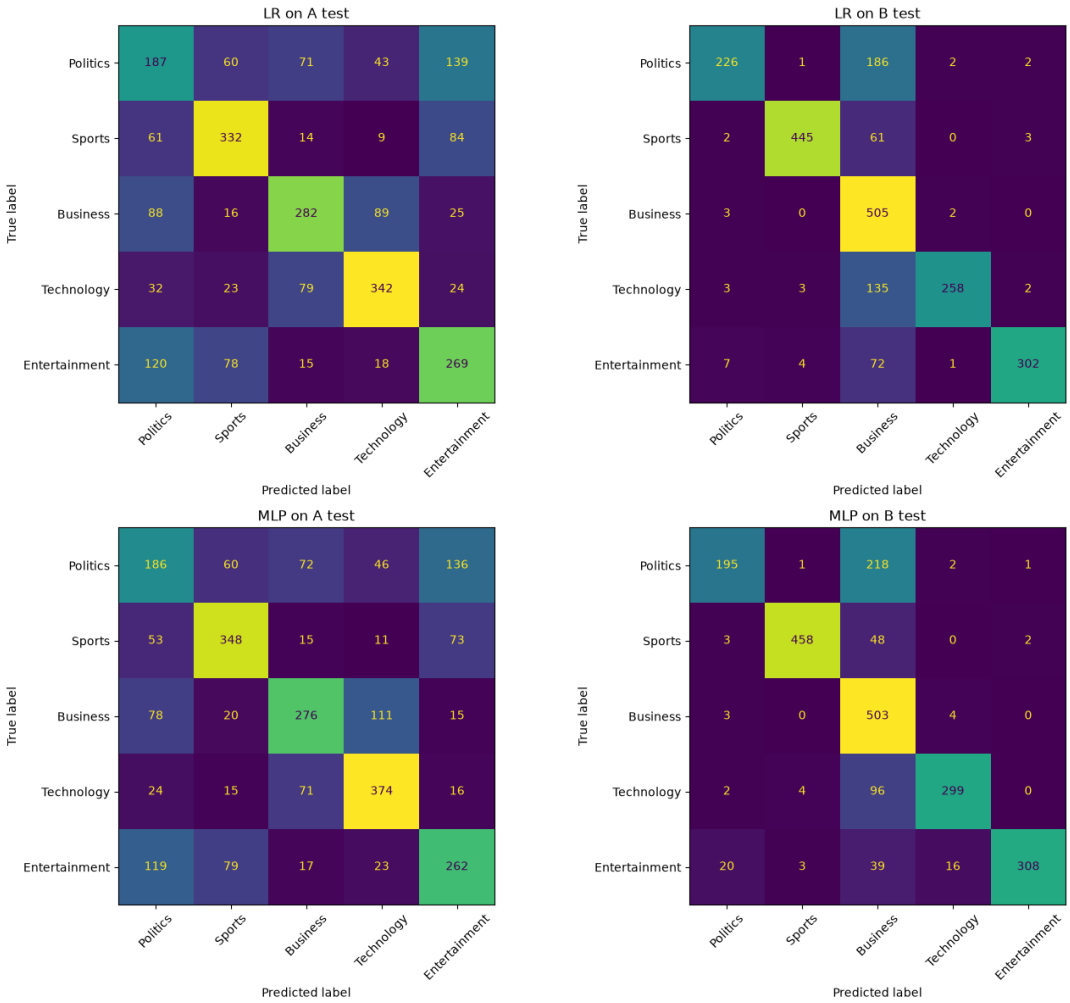


Figure 4. Confusion matrices of LR and MLP for Dataset A and B

Sports did the best for both models. The MLP obtained a macro-F1 of 0.938 on Sports for Data B and LR obtained 0.923. Entertainment and Technology also worked well with both models. Politics remained to be the most difficult category on Data B when it came to F1 and recall. Then for Business both precision and Data A/B's macro-F1 score were extremely low. This means that the model recognized most Business articles but somehow incorrectly assigned the Business label to articles from other classes. It's possible that the class names had different writing styles which led to different ways of classification. For example, Sports would have explicit terms like team names or cities which makes it easier to classify compared to trying to differentiate between Business, Politics, and Entertainment. There was an example involving Star Wars that was labeled as Technology in the raw data when it should've been labeled as Entertainment. Class balance might make it such that the data is able to learn more about one class than another, but the data had an almost even number of examples for each class. Greater text length also gives more information for the model to learn from. When comparing regularization settings, there didn't seem to be much change. For logistic regression, the L2 regularization with a weight decay of 10^{-5} had the best Data B macro-F1 among the other settings at 0.7888 however the weight decay of 10^{-4} decreased Dataset B's macro-F1 to only 0.7597 and the stronger regularization decreased it to 0.5494. It seems like weaker regularization made the model better. As for the MLP, the moderate dropout model achieved a Data B macro-F1 of 0.7941. However, the MLP without regularization achieved a higher Dataset B macro-F1 of 0.823. Then adding dropout and L2 weight decay reduce Dataset B's macro-F1 to 0.7797 while strong dropout and stronger L2 regularization reduced the macro-F1 to 0.7349. Therefore, the results don't support the idea that stronger regularization improves the transfer to the Dataset B test set and Dataset A test set. Table 8 shows the final LR and MLP models that were chosen since they both achieved the highest macro-F1.

TABLE 8. FINAL CONFIGURATIONS FOR LR AND MLP

Model	Model Name	Macro-F1 (A)	Best Epoch	Learning rate	L2	Dropout
LR	`Lr-wdle-5`	0.8013	6	10^{-2}	10^{-5}	None
MLP	`Mlp_dropout`	0.8045	2	10^{-3}	0	None

5. Part 4: Cross-Dataset Transfer to Dataset C

We applied two frozen Part 3 models directly to Data C testing set, which has 2,000 articles of which 500 are in each of labels 1 through 4 evenly. The Part 3 TF-IDF vectorizer was only used with transform, so Data C articles are being represented using the Data A vocabulary that has 19,834 terms and neither model was retrained. As an in-domain reference, we trained separate models on the Data C training set but they were never combined with the transferred models. We also checked to ensure no Data C test set articles were identical to any of the Data A train set articles nor Data C train set articles and the check showed that there were indeed no overlapping articles. The first model we used is the logistic regression model with a single linear layer with 5 outputs trained at learning rate 0.01, weight decay 1e-5 and batch size 64 for 6 epochs. The second is the MLP with two hidden layers of 128 and 64 units with ReLU activations and dropout of 0.2 and 0.3 and trained with a learning rate 0.001 with no weight decay and batch size 64 for 2 epochs. Both of these models were trained on all 7,500 articles in Data A training set with seed 0.

For the labels, sports, business, and technology mean the same thing in Data A and Data C so we analyzed the 1,500 test articles with true labels of 2, 3, and 4. We computed macro-F1 for those three labels where a prediction of 1 or 5 counted as an error. LR reached 0.587 accuracy and 0.629 macro-F1. The MLP reached 0.599 accuracy and 0.638 macro-F1. By analyzing the results, sports transferred well with an F1 of 0.812 on LR and 0.819 on MLP. This is likely because the vocabulary of team names and scores are distinct compared to the other categories. Business and Technology on the other hand were much weaker with F1 between 0.46 and 0.48. This first figure shows that they are confused with each other far more than with any other class. LR labelled 138 Business articles as Technology and 165 Technology articles as Business. The MLP did the same with 156 and 171. Data C business and technology new share a lot of company and market vocabulary that Data A didn't separate well.

If we compare this to Data B, the shared label transfer to Data C is weaker than Data B in part 3 which had a macro-F1 of 0.79. It was also interesting to see that categories with the same meaning still lost accuracy as the source and writing style changed.



Figure 5. Confusion matrices of LR and MLP trained on Dataset A

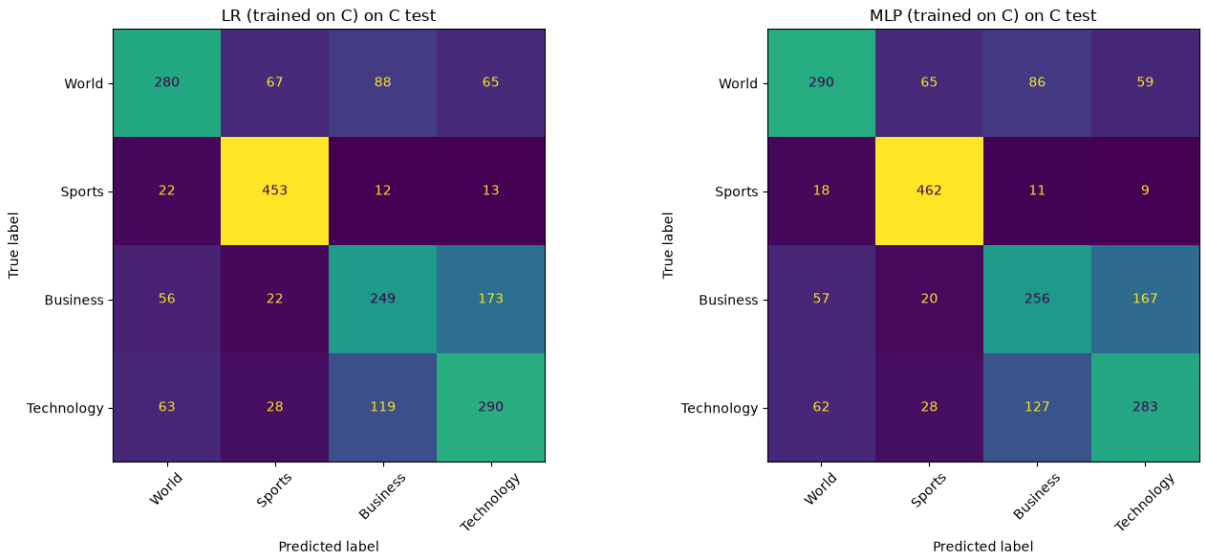


Figure 6. Confusion matrices of LR and MLP trained on Dataset C

Comparing Data A and Data C, the semantic transfer of label 1 changed from Politics in Data A to the broader category World. LR precision became 0.592, recall became 0.430, and F1 became 0.498. Similarly, MLP precision became 0.607, recall became 0.392, and F1 became 0.476. From the 500 true World articles, LR predicted 215 as label 1 and MLP predicted 196. From the plots, whenever World articles were missed, they were often mistaken as Business. LR also labelled 148 non-world articles as World and MLP did that for 127. Label 5 doesn't exist in Data C, so every prediction for that label is wrong. LR made that error for 139 articles and MLP did that for 110 articles. Often it pushed articles about culture, religion, and other stories into entertainment. LR and MLP misclassifications of World articles fell into six groups: foreign conflict and politics, sports/noise, international economics, disasters and society, culture and religion, and a single example of a misclassified tech article. The most common words absent from Data A that led to missed classifications primarily came from terminology about countries like EU, India, and Nigeria or economic terms like prices and crude oil, as well as some international sports terminologies like Olympics and games. This highlights that Data A primarily had articles on US based politics rather than international politics. We also noticed that terms like "afp" and "quot" showed formatting differences between the datasets. As an in-domain reference, our models were trained on Data C with the same Part 3 settings of an 80/20 split of the data, TF-IDF vectorizer on 17303 terms rather than the original 19834 terms, and early epoch stopping on the macro-F1 validations. These models were trained on Data C testing set and as a result never predicted label 5 as they were never exposed to that label in training. This change allowed us to see the true impact of changing label 1 from Politics to World resulting in a F1 of 0.379 for LR and 0.388 for MLP.

TABLE 9. PER-CLASS PERFORMANCES OF LR AND MLP ON TEST DATASET A AND C

Model	Source	Label 1 Recall	Label 1 F1	Label 1–4 Macro-F1
LR	Trained on A (Transfer)	0.430	0.4983	0.5608
LR	Trained on C (Reference)	0.560	0.6080	0.6316
MLP	Trained on A (Transfer)	0.392	0.4763	0.5597
MLP	Trained on C (Reference)	0.580	0.6257	0.6409

Table 9 shows the results of the reference models on Data C testing set comparing whether it was trained on the Data A training set or the Data C training set. From this data, we can see that World F1 scores grew by 11 points for LR and 15 points for MLP showing our estimated tradeoff for Politics and World mismatches in the data. However, there are still challenges with labelling World articles as the topic in Data C is so broad that many of the other articles have overlapping terminology. We can conclude from Part 4 that because LR and MLP stay within a few points of each other in every comparison, the nonlinear model doesn't help with category boundaries which makes sense as it hadn't seen that specific category during training.

6. Part 5: Limited-Data Adaptation to Dataset C

In this part, we started from the final Part 3 models and added a limited number of Data C training articles to see how much a small amount of target-domain data helps. Both models kept the settings selected in Part 3, which were chosen

TABLE 10. LOGISTIC REGRESSION RESULTS BY STRATEGY AND BUDGET

Strategy	Added	C label-1 recall	C label-1 F1	C macro-F1 (1–4)	C strict macro-F1 (2–4)	A test macro-F1	B test macro-F1
Baseline	0	0.430	0.498	0.561	0.582	0.565	0.789
Label 1 only	50	0.552	0.541	0.566	0.574	0.562	0.801
Label 1 only	150	0.630	0.551	0.561	0.564	0.559	0.806
Label 1 only	600	0.714	0.554	0.545	0.542	0.551	0.818
Label 1 only	1,500	0.758	0.540	0.530	0.526	0.549	0.797
Shared only	50	0.408	0.490	0.555	0.577	0.566	0.797
Shared only	150	0.370	0.463	0.553	0.583	0.567	0.803
Shared only	600	0.306	0.420	0.568	0.617	0.569	0.811
Shared only	1,500	0.246	0.363	0.561	0.627	0.575	0.806
Balanced	50	0.460	0.517	0.558	0.572	0.565	0.799
Balanced	150	0.500	0.548	0.579	0.589	0.566	0.805
Balanced	600	0.544	0.581	0.611	0.621	0.571	0.830
Balanced	1,500	0.548	0.584	0.620	0.632	0.570	0.855

TABLE 11. MLP RESULTS BY STRATEGY AND BUDGET

Strategy	Added	C label-1 recall	C label-1 F1	C macro-F1 (1–4)	C strict macro-F1 (2–4)	A test macro-F1	B test macro-F1
Baseline	0	0.392	0.476	0.560	0.588	0.575	0.794
Label 1 only	50	0.556	0.541	0.570	0.579	0.570	0.838
Label 1 only	150	0.584	0.544	0.543	0.543	0.578	0.769
Label 1 only	600	0.716	0.559	0.553	0.551	0.561	0.840
Label 1 only	1,500	0.742	0.559	0.517	0.503	0.565	0.788
Shared only	50	0.424	0.499	0.578	0.604	0.570	0.825
Shared only	150	0.354	0.452	0.557	0.592	0.565	0.809
Shared only	600	0.354	0.460	0.576	0.615	0.567	0.833
Shared only	1,500	0.252	0.368	0.569	0.636	0.572	0.818
Balanced	50	0.430	0.504	0.565	0.586	0.584	0.829
Balanced	150	0.482	0.540	0.592	0.610	0.576	0.806
Balanced	600	0.544	0.580	0.610	0.620	0.574	0.841
Balanced	1,500	0.526	0.581	0.623	0.637	0.586	0.858

without any Data C test results. Logistic regression used a learning rate of 0.01 and weight decay of $1e-5$ for 6 epochs. The MLP used a learning rate of 0.001, no weight decay, and dropout of 0.2 and 0.3 for 2 epochs. Both models used Adam with a batch size of 64, unigram TF-IDF features on the title and description, and random seed 0. Every added article came from the Dataset C train set. The Data C test set was only used for evaluation. We tested budgets of 50, 150, 600, and $1,500$ added articles with three sampling strategies. The label 1 only strategy adds Data C World articles. The shared labels only strategy adds Sports, Business, and Technology articles, split as evenly as possible, so a budget of $1,500$ adds 500 from each class. The balanced strategy adds about the same number from all four classes, which gives 375 per class at $1,500$. To make the sampling reproducible, each Data C class was shuffled once with seed 0, and every sample takes the first n articles from each shuffled class. Because of this, a smaller sample is always contained in a larger one. For every condition, we refit the TF-IDF vectorizer and the classifier on Data A train plus the sampled Data C articles, so the training set grew from $7,500$ articles to as many as $9,000$. The zero-example baseline is the frozen Part 3 model, and it was run once per model.

As shown in Tables 10 and 11, the balanced strategy gave the best balance between improving on Data C and keeping performance on the original data. It was the only strategy where Data C macro-F1 kept going up with every budget. At $1,500$ added articles, logistic regression went from 0.561 to 0.620 and the MLP went from 0.560 to 0.623. These scores come close to the in-domain reference models from Part 4, which reached 0.632 for LR and 0.641 for the MLP after training on $4,800$ Data C articles. Data A macro-F1 stayed within about a point of the baseline for both models, and Data B improved, rising from 0.789 to 0.855 for LR and from 0.794 to 0.858 for the MLP. Adding only World articles raised World recall by a lot. LR recall went from 0.430 to 0.758 and MLP recall went from 0.392 to 0.742. World F1 only rose to about 0.54 to 0.56, though, because precision dropped as the models started labeling many Sports, Business, and Technology articles as World. By $1,500$ articles, overall Data C macro-F1 fell below the baseline to 0.530 for LR and 0.517 for the MLP, and strict macro-F1 on labels 2 to 4 dropped as well. The MLP's Data B score was also unstable with this strategy, falling to 0.769 at 150 articles before recovering. The shared labels only strategy had the opposite problem. It gave the highest strict macro-F1 on labels 2 to 4 at most budgets, reaching 0.627 for LR and 0.636 for the MLP at $1,500$. World performance got worse with every budget, though. World recall fell to about 0.25 for both models, and World F1 dropped to 0.363 for LR and 0.368 for the MLP. Since the models saw more examples of the other three classes and none of World, they had even less reason to predict World than before. Overall Data C macro-F1 barely changed from the baseline. These results show that label-1 examples are necessary. Without them, the models got worse at recognizing World articles, which is the main source of error when moving from Data A to Data C. World examples alone are not enough either, since they teach the

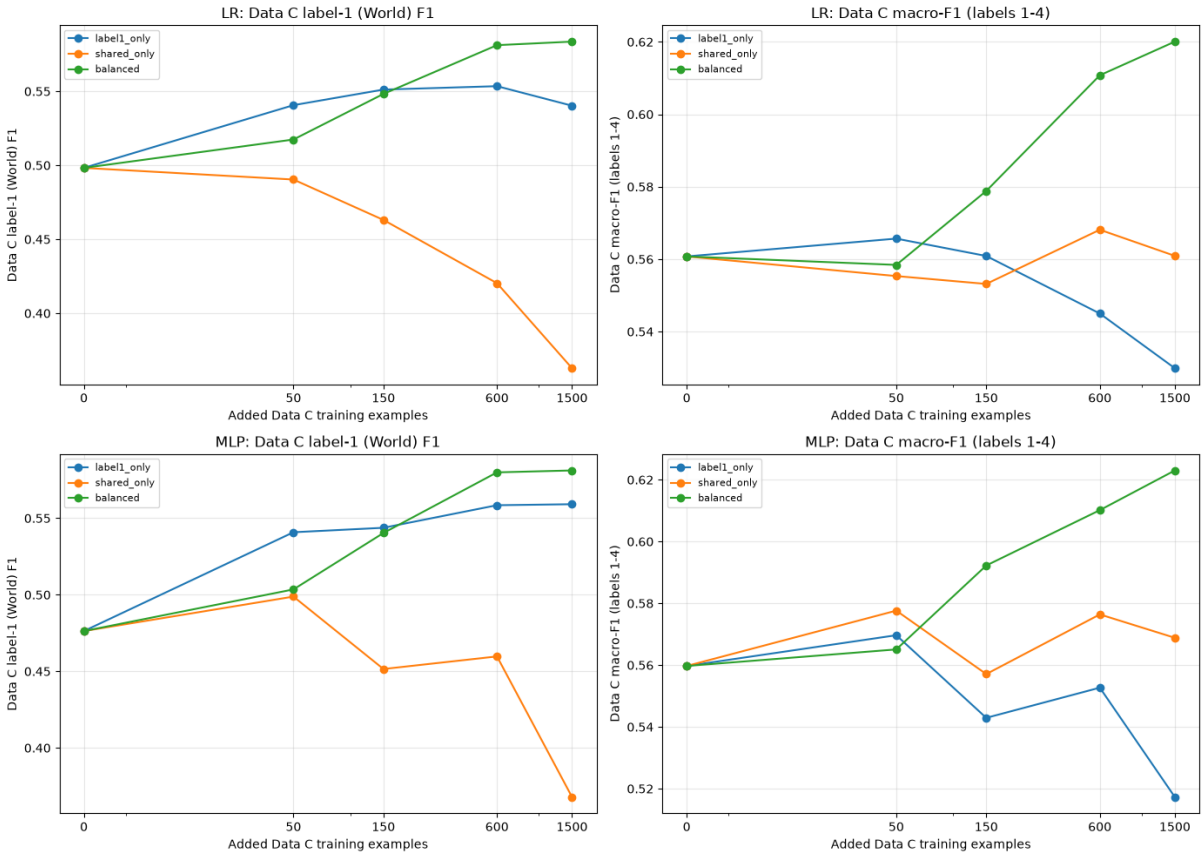


Figure 7. Data C label-1 (World) F1 and Data C macro-F1 as Data C training examples are added, for logistic regression (top) and the MLP (bottom).

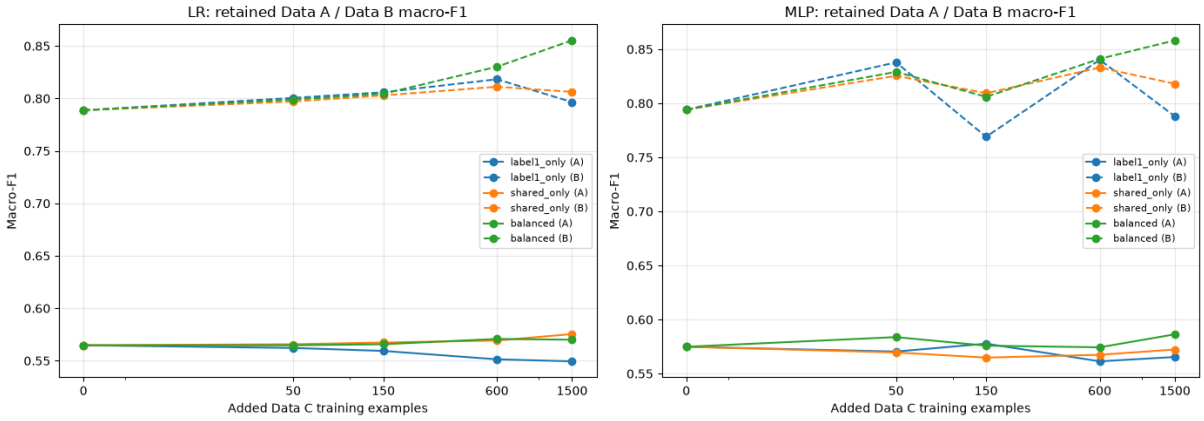


Figure 8. Data A test macro-F1 (solid) and Data B test macro-F1 (dashed) as Data C training examples are added, for logistic regression (left) and the MLP (right).

model what World looks like without showing where it ends. Balanced contrast examples help the model separate World from Sports, Business, and Technology. At $1,500$ articles, the balanced strategy reached a higher World F1 than the label 1 only strategy for both models, even though it added only 375 World articles instead of $1,500$. It also matched the strict macro-F1 of the shared labels only strategy, with 0.632 compared to 0.627 for LR and 0.637 compared to 0.636 for the MLP. The balanced models improved on World and the shared labels at the same time, which neither of the other strategies managed. All 24 adapted models passed the 40% macro-F1 requirement on Data C. The lowest was the MLP with $1,500$ label 1 only articles at 0.517.

7. Part 6: Text-Representation Extension

In the previous sections, we analyzed the performance of different models using datasets represented with unigram TF-IDF features. In this section, we extend the text representation by using bigrams in addition to unigrams and examine how the resulting increase in feature dimensionality affects model performance. The experiments are designed to answer the following questions: (a) How do the models respond to the increased feature dimensionality introduced by bigram features? (b) Does the unigram–bigram representation improve both in-domain performance and cross-dataset transfer compared with the unigram baseline? and (c) Does the extended representation improve model adaptation to Dataset C when limited target-domain training examples are added?

TABLE 12. VOCABULARY SIZES IN DIFFERENT TEXT REPRESENTATION POLICY

Representation	Split vocabulary	Full-training vocabularly
Unigram TF-IDF	$17,857$	$19,834$
Unigram + bigram	$25,465$	$31,068$

As shown in Table 13, the unigram produces a vocabulary of $19,834$ features when fitted on the full Dataset A training set. By using both unigrams and bigrams with `min_df=2`, the vocabulary increases to $31,068$ features, corresponding to an increase of approximately 56.64%. Bigrams allow the representation to preserve short two-word pairs that may carry more specific semantic information than individual words. For example, phrases such as "prime minister" or "stock market" may provide stronger class-related information than the individual terms considered separately. Note that the extension changes only the text representation; the data splits, random seed, model hyperparameters, and evaluation metrics remain unchanged. This controlled comparison isolates the effect of feature representation as far as possible. The final LR configuration uses unigram–bigram TF-IDF with `min_df=2`, Adam, learning rate of 0.01, $\mathcal{L}2$ regularization of 10^{-5}, epochs of 11, and batch size of 64. The final MLP uses $128 \rightarrow 64$ ReLU hidden units and dropout layers with the rates of $0.2 \rightarrow 0.3$, Adam with learning rate of 0.001, epochs of 8, and batch size of 64. After refitting on all $7,500$ Dataset A training set, both models run on a $31,068$ vocabularies.

We first compare the extended models with their unigram counterparts on Datasets A and B to examine in-domain and cross-source performance. We then repeat the limited-data adaptation experiments on Dataset C to determine whether the additional phrase-level features improve adaptation under World-only, shared-label-only, and balanced sampling strategies.

TABLE 13. ACCURACIES AND MACRO-F1 SCORES OF UNIGRAM/BIGRAM LR AND MLP ON DATASET A AND B

Model	Accuracy on Dataset A	Macro-F1 on Dataset A	Accuracy on Dataset B	Macro-F1 on Dataset B
Unigram LR	.5648	.5647	.7802	.7888
Bigram LR	.5672	.5667	.7146	.7255
Unigram MLP	.5784	.5749	.7924	.7941
Bigram MLP	.5720	.5715	.7924	.7988

On Dataset A test set, LR macro-F1 increases from 0.5647 to 0.5667, while MLP macro-F1 decreases from 0.5749 to 0.5715. The cross-source Dataset B results are more informative. LR macro-F1 falls from 0.788 to 0.7255 (-0.0633), whereas the MLP increases slightly from 0.7941 to 0.7988 ($+0.0047$). Therefore, the larger feature space does not necessarily produce a consistent transfer across different models.

TABLE 14. PER-CLASS PRECISION, RECALL, AND F1 SCORES OF BIGRAM LR AND MLP ON DATASET A AND B

Model	Class	Precision (A)	Recall (A)	F1 (A)	Precision (B)	Recall (B)	F1 (B)
Bigram LR	Politics	.395	.396	.396	.917	.480	.630
Bigram LR	Sports	.645	.670	.658	.976	.787	.871
Bigram LR	Business	.582	.554	.568	.459	.990	.627
Bigram LR	Technology	.676	.688	.682	.976	.509	.669
Bigram LR	Entertainment	.533	.528	.531	.976	.723	.830
Bigram MLP	Politics	.388	.384	.386	.934	.542	.686
Bigram MLP	Sports	.673	.686	.679	.978	.885	.929
Bigram MLP	Business	.577	.548	.562	.546	.984	.703
Bigram MLP	Technology	.673	.680	.677	.968	.668	.791
Bigram MLP	Entertainment	.546	.562	.554	.969	.816	.886

The class-wise results further illustrate this difference. In Table 14, F1 score of LR decreases fro all five classes after adding bigrams, with the largest reduction occurring for Technology ($0.777 \rightarrow 0.669$). For the MLP, Politics improves from 0.6094 to 0.686 and Entertainment changes only slightly from 0.884 to 0.886, while Sports, Business and Technology decline. These changes suggest that proximity-token features can be informative for particular categories without being uniformly transferable across the full label set.

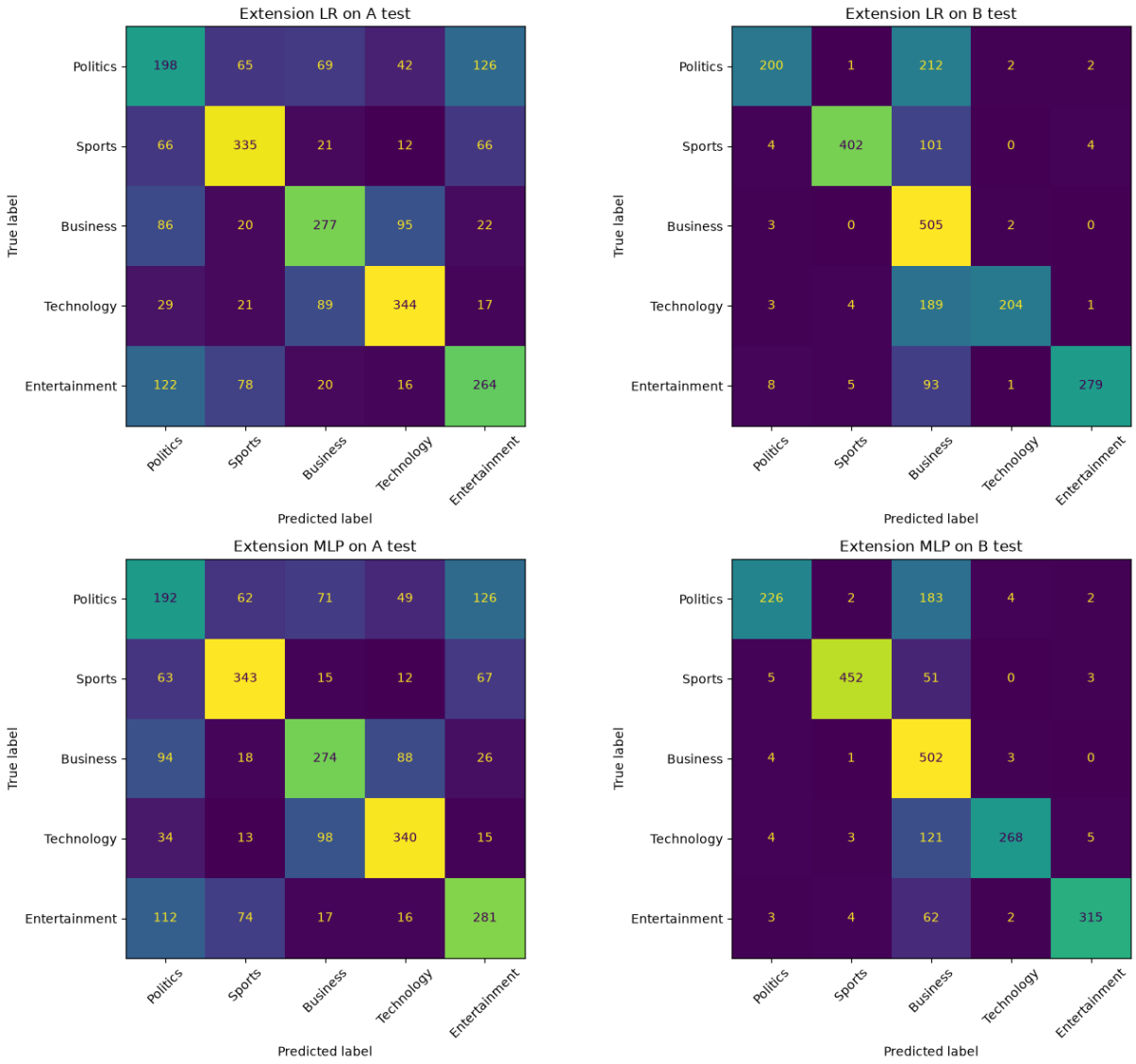


Figure 9. CConfusion matrices of extended LR and MLP on Test Dataset A and B

The confusion matrices provide a concrete explanation for the mixed transfer effects. The extended LR produces additional false Business predictions on Dataset B, including Politics–Business, Sports–Business, and Technology–Business errors. This is consistent with the possibility that some two-word pairs associated with Business in Dataset A are also present in other categories from Dataset B. In contrast, the MLP can combine unigram and bigram evidence nonlinearly and obtains a small improvement, although that improvement is not shared by every class. The extension therefore changes the error structure rather than simply increasing or decreasing performance uniformly.

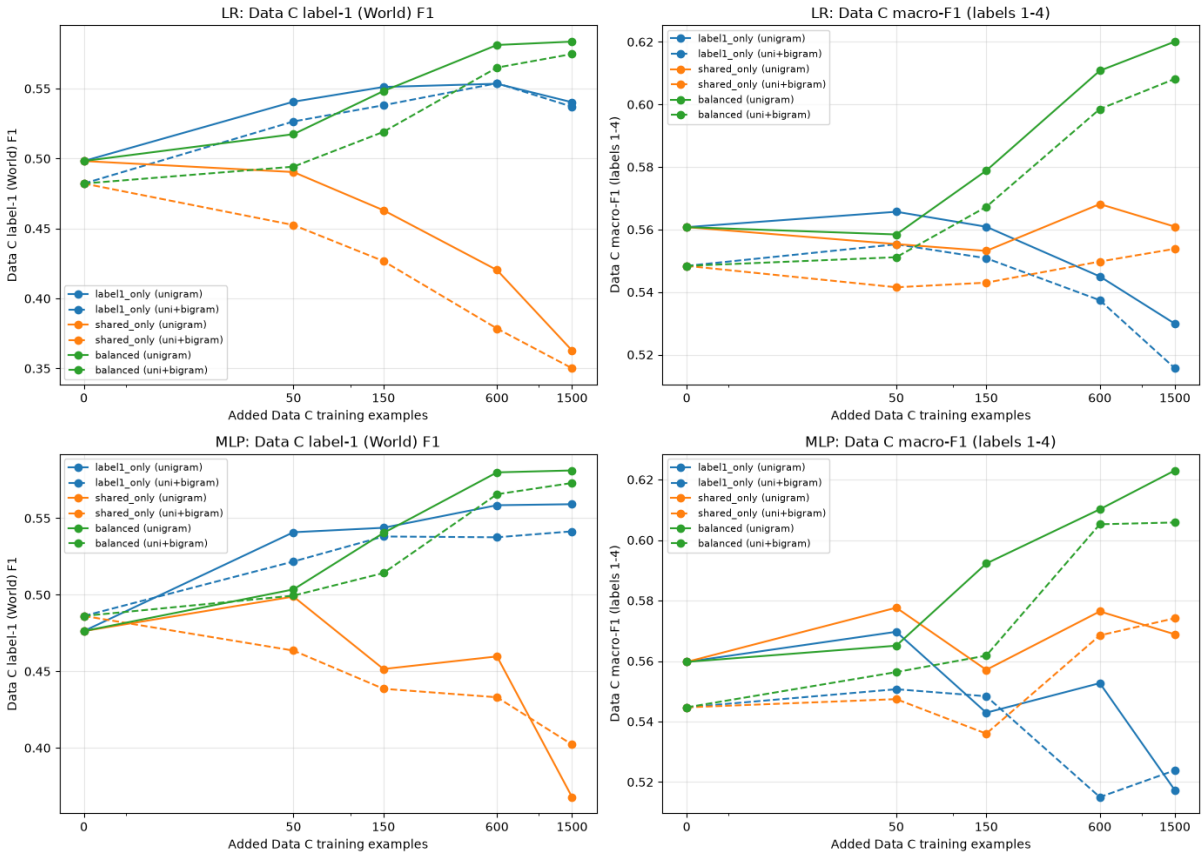


Figure 10. LR and MLP Performance under Different Data C Training Strategies and Feature Representations

8. Conclusion

In this work, we systematically analyzed the generalization and adaptation performance of custom MLP models under different settings and hyperparameters. Our experiments demonstrated the MLP's generalization ability in three different settings: in-domain performance on Dataset A was limited by topical overlap and source-specific patterns; cross-source transfer to Dataset B achieved strong overall performance while exhibiting noticeable changes in class-wise precision and recall; and the semantic category shift in Dataset C revealed limitations in transferring a decision boundary to a dataset with a different class composition. The adaptation experiments provided the clearest evidence of the effects of category mismatch. World-only sampling increases World recall but reduces performance on the remaining classes, whereas shared-label-only sampling improves performance on non-World classes at the expense of World recall. Balanced sampling mitigates both effects by adding World examples along with examples from the other classes, helping the model classify World while preserving its ability to distinguish among the remaining categories. In a further analysis, we showed that increasing feature dimensionality does not necessarily improve transfer performance. Using bigrams increases the full-training vocabulary size by 56.64% but generates inconsistency in performance improvements before adaptation to Dataset C and, overall, underperform the corresponding unigram models. Based on these results, we conclude that unigram TF-IDF provides the more effective and consistent text representation for the final models.